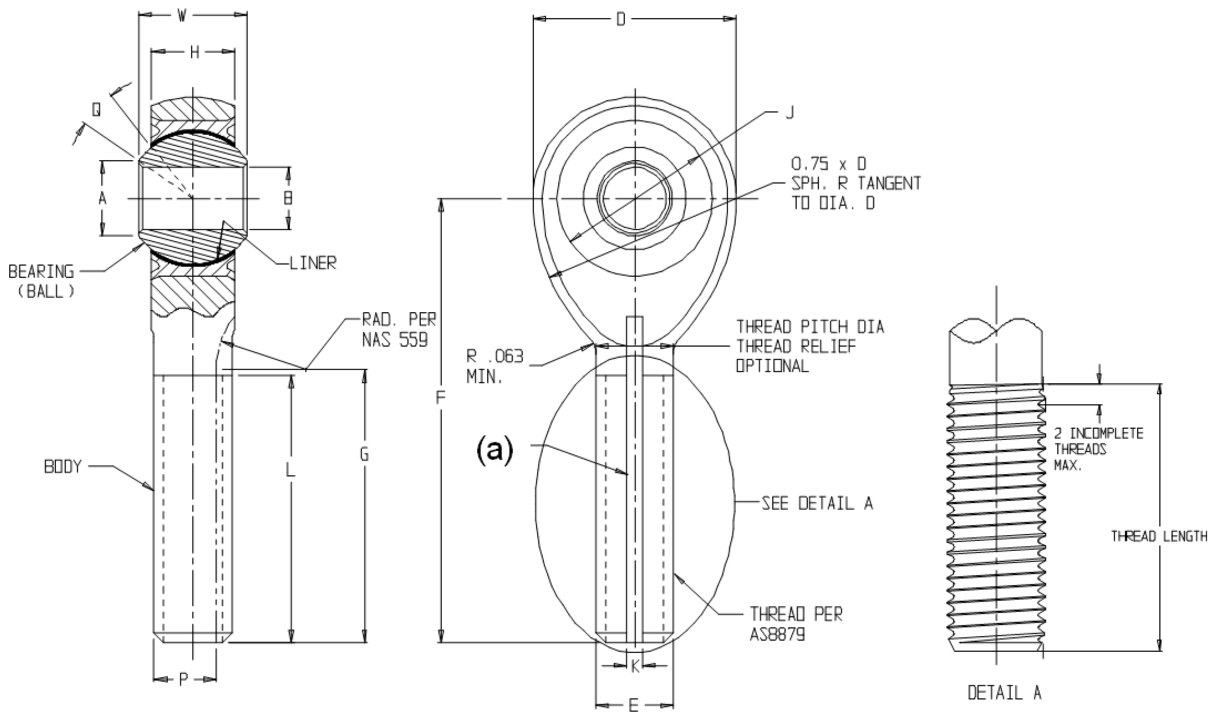
RATIONALE

THIS IS THE INITIAL RELEASE OF AN ALL CORROSION RESISTANT STEEL (CRES) VERSION OF AS81935/8.

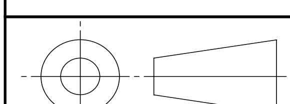


(a) KEYWAY SHALL BE ORIENTED ALONG THE CENTERLINE OF THE THREADED SHANK. THE PLANE OF THE BOTTOM OF THE KEYWAY SHALL BE PARALLEL TO THE PLANE MADE BY THE FACE OF THE ROD END HOOP.

FIGURE 1 – ROD END ASSEMBLY

ISSUED 2011-06

THIRD ANGLE PROJECTION



CUSTODIAN: ACBG – PLAIN BEARING

PROCUREMENT SPECIFICATION: AS81935

SAE Aerospace
An SAE International Group

AEROSPACE STANDARD

BEARING, PLAIN, ROD END, SELF-ALIGNING, SELF-LUBRICATING, NARROW, EXTERNALLY THREADED, STEEL, CORROSION RESISTANT (CRES)-65 TO +325 °F

SAE AS81935/8

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Fax: 724-776-0790 **Email: CustomerService@sae.org** **SAE WEB ADDRESS: http://www.sae.org**

TABLE 1 - DIMENSIONS

DASH NUMBER	B BORE +.0000 -.0005	D SPH. DIA. ± .010	L THREAD LENGTH ±.031	E THREAD SIZE UNJF-3A (c)	F CTRLINE BALL TO END ±.010	W BALL WIDTH +.000 -.002
-03	.1900	.680	.775	.2500-28	1.315	.281
-04	.2500	.827	.775	.2500-28	1.443	.343
-05	.3125	.984	1.187	.3125-24	1.948	.375
-06	.3750	1.131	1.187	.3750-24	2.030	.406
-07	.4375	1.294	1.281	.4375-20	2.250	.437
-08	.5000	1.459	1.468	.5000-20	2.544	.500
-10	.6250	1.763	1.562	.6250-18	2.832	.625
-12	.7500	2.140	1.687	.7500-16	3.193	.750
-14	.8750	2.372	2.000	.8750-14	3.677	.875
-16	1.0000	2.681	2.100	1.0000-12	3.968	1.000

DASH NUMBER	H BODY WIDTH ± .005	A MIN	J MAX HOUSING I.D.	Q° MIN	G KEYWAY FLAT	K KEYWAY (b) +.005 -.000	P +.000 -.005
-03	.228	.293	.5625	10	.896/.836	.062	.207
-04	.260	.364	.6562	10	.896/.876	.062	.207
-05	.291	.419	.7500	10	1.308/1.288	.062	.268
-06	.322	.475	.8125	9	1.308/1.288	.093	.319
-07	.353	.530	.9062	8	1.402/1.382	.093	.383
-08	.400	.600	1.0000	8	1.589/1.569	.093	.445
-10	.510	.739	1.1875	8	1.683/1.663	.125	.541
-12	.603	.920	1.4375	8	1.808/1.788	.125	.663
-14	.713	.980	1.5625	8	2.121/2.101	.156	.777
-16	.807	1.118	1.7500	9	2.221/2.201	.156	.900

(b) KEYWAY, WHEN SPECIFIED, IS COMPATIBLE WITH LOCKING DEVICES: AS81935/3 FOR SIZES –03 THROUGH –08, AND NAS 513 FOR SIZES –10 THROUGH –16. KEYWAY TOLERANCES NOT SPECIFIED SHALL BE IN ACCORDANCE WITH AS81935/3 OR NAS 513 AS APPLICABLE.

(c) THREADS SHALL BE FORMED BY A SINGLE ROLLING OPERATION AFTER FINAL HEAT TREAT AS SPECIFIED IN THE THREAD SECTION OF AS81935.

TABLE 2 – LOAD VALUES

DASH NUMBER	(d) ULTIMATE STATIC LOAD (LBS)	(e) FATIGUE LOAD (LBS)	AXIAL PROOF LOAD (LBS)	WEIGHT REF (LBS)	NO LOAD ROTATIONAL BREAKAWAY TORQUE (IN-LBS) MIN	 MAX
-03	3000	1100	150	.045	.5	6
-04	5300	1500	430	.060	.5	6
-05	8600	2400	700	.100	1	15
-06	13 000	3600	1100	.135	1	15
-07	17 800	5000	1400	.200	1	15
-08	24 200	6800	2040	.285	1	15
-10	38 500	10 800	2430	.505	1	15
-12	56 600	16 000	2940	.830	1	15
-14	77 400	21 900	3190	1.235	1	24
-16	101 400	28 600	3570	1.725	1	24

(d) ULTIMATE LOADS ARE ANALYTICAL VALUES BASED ON ROD END BANJO FOR –03 AND THREADS FOR ALL OTHER SIZES.

(e) FATIGUE LOADS ARE ANALYTICAL VALUES BASED ON 50 000 CYCLE LIFE. FATIGUE LOADS ARE DEFINED PER PROCUREMENT DOCUMENT AS81935.

REQUIREMENTS:

1. MATERIAL:

   BODY: PH13-8MO CRES PER AMS5629

   BEARING CARTRIDGE: MS14101-XX WITH A 440C AMS5630 BALL SHALL BE USED AS THE DEFAULT BEARING INSERT. MS14101-XXC WITH A PH13-8MO AMS5629 BALL SHALL BE USED WHEN SPECIFIED IN THE PART NUMBER FOR THE ROD END ASSEMBLY. MS14101A-XX BEARING SHALL BE USED WHEN SPECIFIED IN PART NUMBER. SUFFIX “A” INDICATES THE BEARING CARTRIDGE UTILIZES A PTFE LINER THAT IS AS81820 TYPE “A” QUALIFIED”. ABSENSE OF “A” INDICATES STANDARD PTFE LINER QUALIFIED TO AS81820. SEE BELOW FOR PART NUMBER EXAMPLE.

2. HARDNESS:

   BODY: HRC 40 TO 44, HEAT TREAT TO CONDITION H1050, IN ACCORDANCE WITH AMS-H-6875 OR AMS2759/3

3. SURFACE TEXTURE: PER ANSI B46.1

   BODY: BORE, Ra 32 MAX., SIDES OF THE THREAD AND ROOT AREA, Ra 32, THREAD RELIEF, Ra 63.
   ALL OTHER MACHINED SURFACES, Ra 125 MAX.

4. CLEANING: PASSIVATE ROD END BODY AS DETAIL PER AMS2700, METHOD 1 (NITRIC ACID) OR METHOD 2 (CITRIC ACID), OR ASTM-A-967 CITRIC 1, CITRIC 2, AND CITRIC 3 ARE ACCEPTABLE.

5. TOLERANCES: UNLESS OTHERWISE SPECIFIED, DECIMALS ±.010, ANGLES ± ½ DEGREE.

6. DIMENSIONS IN INCHES, UNLESS OTHERWISE SPECIFIED.

7. BREAK SHARP EDGES AND CORNERS AND REMOVE ALL BURRS AND SLIVERS.

8. NONDESTRUCTIVE TESTING: THE ROD END BODY ONLY, AFTER ALL FINISH MACHINING, HEAT TREAT, PASSIVATION, AND PRIOR TO BEARING INSTALLATION, SHALL BE MAGNETIC PARTICLE INSPECTED PER ASTM-E-1444. THE ACCEPTANCE CRITERIA SHALL BE AS DEFINED IN THE PROCUREMENT DOCUMENT AS81935 (UNDER MAGNETIC PARTICLE INSPECTION).

9. NOTE: THE PARTS DEFINED IN THE AS81935/8 DOCUMENT ARE MARKED WITH M81935/8-XXXX; SEE EXAMPLE BELOW.

10. EXAMPLE OF PART NUMBER:

M81935/8 - 04 C K L A = .2500 NOMINAL BORE DIAMETER, PH13-8MO BALL MATERIAL, WITH KEYWAY, LEFT HANDED THREAD, BEARING CARTRIDGE WITH TYPE “A” LINER,

- A: ABSENSE OF LETTER INDICATES STANDARD PTFE LINER QUALIFIED TO AS81820, “A” INDICATES THE BEARING CARTRIDGE UTILIZES A PTFE LINER THAT IS AS81820 TYPE “A” QUALIFIED
- L: ABSENCE OF LETTER INDICATES RIGHT HANDED THREAD
  LETTER “L” INDICATES LEFT HANDED THREAD
- K: ABSENCE OF LETTER INDICATES NO KEYWAY
  LETTER “K” INDICATES KEYWAY. “W” INDICATES DEEP SQUARE SLOT KEYWAY.
- C: ABSENCE OF LETTER INDICATES 440C BALL OPTION
  LETTER “C” INDICATES PH13-8MO BALL OPTION
- 04: DASH NUMBER CITING THE BORE NOMINAL DIAMETER CODE (2 DIGITS REQUIRED)
- M81935/8: BASIC PART NUMBER